Analysis of Embryo Cell Trajectory with Co-clustering Techniques

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Introduction to Embryo Cell Trajectory Analysis

- Objective: to analyze the trajectory of cells in an embryo and its relationship with cell fate or cell category.
- Importance: highlight the importance of understanding cell trajectories for advancing developmental biology and potential medical applications.

Methodology Overview

- **Stochastic Modeling:** e.g. Drift-diffusion model, Heston model, etc.
- **Classification and Analysis:** the stochastic models can reveal an expected trajectory pattern and variance characteristics, which can be used to classify embryos and cells based on observable trajectories.

Feature Analysis

- **Cell category:** e.g. Average velocity, average pressure, average stress, etc.
- **Fate transition:** e.g. Variance of velocity, variance of pressure, variance of stress, etc.

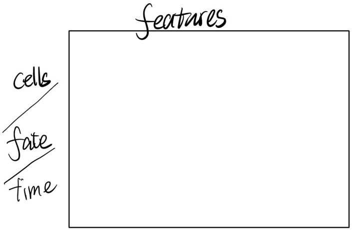

Figure: Cell trajectory

Conclusions and Discoveries

- **Cell prospective:** e.g. Cell of category A tends to have a trajectory with a higher winding number, etc.
- **Time/Fate prospective:** e.g. During the phase changing, the variance of the trajectory tends be maximum, etc.